

SATADEEP DASGUPTA

Senior Software Engineer

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> Professional Summary

Systems engineer with 5+ years building production distributed systems, workflow runtimes, and AI infrastructure using Rust, Python, and TypeScript. Built AIME — an AI-powered education ecosystem with a distilled 2B reasoning model (offline CPU inference), a published npm component library (569+ weekly downloads), and a cross-platform Tauri desktop app shipping on 5 platforms. Designed Loom, a durable workflow engine with event-sourced deterministic replay. Published researcher (hybrid RAG — featured on SurrealDB blog). Currently Senior Software Engineer at Brainium, designing systems serving 50,000+ users with 99.9% uptime.

> Experience

Senior Software Engineer

[Brainium Information Technologies Pvt Ltd](#)

June 2024 – Present

Senior Software Engineer (January 2025 – Present)

- ▶ Lead architecture design for enterprise software systems and cloud-native applications serving 50,000+ users
- ▶ Design scalable backend architectures and distributed workflows with 99.9% uptime requirements
- ▶ Develop AI-assisted systems including AIME (Adaptive Instruction & Misconception Engine), presented to national leadership and 14 state education inspectors
- ▶ Built Loom workflow orchestration engine with deterministic execution and event-sourced replay capabilities
- ▶ Guide system design decisions including service decomposition, data architecture, and observability strategies for mission-critical applications

Associate Software Engineer (June 2024 – January 2025)

- ▶ Implemented core system components for enterprise applications handling 100K+ daily transactions
- ▶ Solved complex backend engineering problems involving integrations and data pipelines, reducing processing time by 40%
- ▶ Worked on distributed application components and performance-sensitive backend services with sub-100ms response times
- ▶ Contributed to codebase modernization efforts improving maintainability and reducing technical debt

Full Stack Developer

[IGLOBAL IMPACTS ITES Pvt Ltd](#)

June 2022 – June 2024

- ▶ Designed and maintained scalable web applications and backend services supporting 10,000+ concurrent users
- ▶ Built and maintained CI/CD pipelines reducing deployment time by 50% and deployment errors by 90%

- Led software releases improving application performance by 40% and reducing page load times by 30%
- Participated in technical hiring and engineering evaluation processes, conducting 30+ technical interviews
- Implemented production-grade RESTful APIs and GraphQL services with 99.5% uptime and sub-200ms response times
- Mentored junior developers on best practices in code versioning, testing, and system design

Freelance Software Developer

August 2019 – June 2022

- Delivered 25+ end-to-end web application projects for international clients across 8 countries
- Designed full-stack systems from requirement gathering to deployment with 100% on-time delivery rate
- Built scalable backend APIs serving 50K+ requests per day with optimized frontend achieving 95+ Lighthouse scores
- Maintained production deployments and resolved complex technical issues with average resolution time under 4 hours
- Achieved 5.0/5.0 client satisfaction rating with multiple recurring clients

> Technical Skills

Languages

Rust C++ Python TypeScript JavaScript Java

Backend & APIs

Node.js Django Express.js REST APIs GraphQL gRPC Tauri

Frontend

React Next.js Vue.js Tailwind React Native

Databases

PostgreSQL MongoDB MySQL Redis SurrealDB

DevOps & Tools

Docker AWS CI/CD Git GitHub Actions Jenkins

Specialized

Event Sourcing Workflow Orchestration Distributed Systems Systems Programming Model Distillation
Edge AI / Offline Inference LLM Engineering Microservices

> Key Projects

AIME Ecosystem — AI Education Platform + Toolchain

- **Platform:** AI-powered learning engine generating structured lessons, detecting misconceptions in real-time, with Lesson Runtime Format (LRF) for replayable execution — presented to national leadership and 14 state education inspectors
- **Reasoner-2B:** Distilled 2B reasoning model (Llama 4 + Opus chains) with Think Cache (pre-warmed KV state) and Think Book (offline knowledge bundles) — runs on CPU-only devices, zero internet required
- **Lesson Studio:** Tauri + Rust desktop app shipping native installers across Linux, ChromeOS/ARM64, Windows, and macOS — assessment editing, markdown print, .aimepack portable file format
- **@aime.ai/renderer-react:** Published npm component library (569+ weekly downloads, 13 versions) — 20+ slide types, BlockSuite whiteboard with Yjs persistence, inline editing primitives, full i18n/RTL

- **Loom:** Durable workflow engine with event-sourced deterministic replay for fault-tolerant orchestration of lesson execution
- [Python](#), [Rust](#), [Tauri](#), [React](#), [TypeScript](#), [Model Distillation](#), [Edge AI](#), [gRPC](#), [BlockSuite](#), [Yjs](#)

OxideDB — LLM-Powered Database System

- Dual query modes supporting SQL and natural language queries via LLM integration
- High-performance gRPC protocol with Protocol Buffers for binary communication
- Multi-language SDKs (Rust, Python, JS/TS) with multi-tenant namespace isolation and JWT auth

[Rust](#), [SQLite](#), [LLM](#), [gRPC](#), [Protocol Buffers](#)

Streammint — Real-Time Messaging Infrastructure

- Pluggable Rust backend with sub-millisecond message routing and modular storage/auth/transport layers
- TypeScript React frontend for live stream interaction, media management, and real-time chat
- Horizontally scalable architecture designed for production messaging workloads

[Rust](#), [TypeScript](#), [React](#), [Real-time Systems](#), [WebSocket](#)

Data2BI — Conversational Business Intelligence

- Natural language interface for data querying — non-technical users perform complex analytics conversationally
- AI-driven pattern detection, anomaly identification, and automatic dashboard generation
- Multi-source integration with real-time processing for operational intelligence

[Python](#), [LLM](#), [Data Analytics](#), [Machine Learning](#), [Conversational AI](#)

Ironwatch — DDoS Mitigation System

- Rust-based adaptive traffic monitoring with AbuseIPDB reputation scoring
- Reduced attack traffic by ~80% in production — threshold-based spike detection with automated blocking

[Rust](#), [Network Security](#), [Traffic Analysis](#)

cvxtract — Schema-Driven Document Extraction

- Rust library for structured data extraction from PDF/DOCX/HTML using LLMs with compile-time type guarantees via Serde + JSON Schema
- Provider-agnostic: local inference (llama.cpp) and cloud APIs — direct deserialization into Rust structs

[Rust](#), [LLM](#), [Document Processing](#), [Serde](#), [JSON Schema](#)

ATXFloods — Critical Flood Monitoring System (City of Austin)

- Production emergency system monitoring flood conditions and road closures across Texas — actively used by City of Austin for public safety
- Django + PostgreSQL backend with React frontend, deployed on AWS EC2/CloudFront — handles 100x traffic spikes during flood emergencies
- Real-time alerts, multi-source data integration, and high-availability architecture for critical disaster response

[Python](#), [Django](#), [PostgreSQL](#), [React](#), [AWS EC2](#), [AWS CloudFront](#)

> Key Achievements

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- ★ **AIIME Ecosystem:** Built 5-component AI education platform (model + desktop app + npm library + workflow engine + platform) — presented to national leadership
 - ★ **Published npm Package:** [@aime.ai/renderer-react](#) — 569+ weekly downloads, 13 versions, used in production across African classrooms
 - ★ **Published Research:** Co-authored hybrid RAG paper (18% accuracy gain) — featured on SurrealDB official blog (27K+ GitHub stars)
 - ★ **Cross-Platform Shipping:** AIIME Lesson Studio ships native installers on Linux, ChromeOS/ARM64, Windows, and macOS via Tauri + Rust
 - ★ **Production Systems:** 99.9% uptime, sub-100ms latency, 80% DDoS traffic reduction across deployed infrastructure

★ **Open Source:** 15+ active repositories — Rust systems tooling, LLM infrastructure, developer CLIs

> Research & Publications

Bridging Analytics and Semantics: A Hybrid Database Approach to Retrieval-Augmented Generation

Co-author | Zenodo Repository, September 2025 | [Featured on SurrealDB Official Blog](#)

DOI: [10.5281/zenodo.17018699](https://doi.org/10.5281/zenodo.17018699) | Source: github.com/satadeep3927/hybridrag | Blog: surrealdb.com/blog

- Hybrid retrieval combining SQL-style analytics with semantic vector search in SurrealDB — 18% accuracy improvement over vector-only baselines
- Research highlighted by SurrealDB (27K+ GitHub stars) as community innovation on their official blog

> Education

Bachelor of Science (Mathematics)

Brahmananda Keshab Chandra College

2019 – 2022

> Languages

- English (Fluent)
- Bengali (Native)
- Hindi (Fluent)